Differentially Private Federated Learning on CIFAR-10: An Empirical Evaluation and Reproducibility Analysis

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Abstract

This report evaluates a Differentially Private Federated Learning (DP-FL) system on CIFAR-10 partitioned across five clients under Dirichlet data heterogeneity ($\alpha \in \{0.1, 10.0\}$). Using Flower and Opacus, we examine privacy–utility tradeoffs across noise multipliers ($\sigma \in \{0.5, 1.0, 1.5, 2.0\}$) and clipping thresholds ($C \in \{0.1, 1.0\}$) under add/remove-one-sample Differential Privacy ($\delta = 10^{-5}$). Centralized non-private training achieves 74.87% test accuracy (Adam, 30 epochs), while non-private FedAvg achieves 47.38% (SGD, 15 rounds). Under DP-FL with $\sigma = 1.0, C = 1.0$ ($\alpha = 0.1$, seed 42), the primary 15-round run achieves 26.11% test accuracy at an accountant-reported estimate of $\varepsilon = 2.6974$ under assumed Poisson subsampling. Across three historical runs with nominally identical configurations, final accuracy ranges from 24.41% to 28.74%, representing an observed spread of approximately 4.33 percentage points. In privacy grid evaluations, configuring $\sigma = 2.0$ with $C = 0.1$ attains 30.57% accuracy at an accountant-reported $\varepsilon = 0.9117$. Crucially, centralized and federated baselines differ in optimizer, learning rate, and training budget, precluding attributing the performance gap solely to differential privacy.

1 Introduction and Related Work

Federated Learning (FL) enables distributed edge devices to collaboratively train a shared machine learning model without centralizing raw data [1]. Although raw training data remain on local devices, sharing parameter updates or gradients remains susceptible to inference and data reconstruction attacks that can recover private training instances. Differential Privacy (DP), specifically through Differentially Private Stochastic Gradient Descent (DP-SGD) [2], mitigates these risks by bounding individual gradient contributions and injecting calibrated noise.

In this work, we implement and evaluate a DP-FL system for image classification on CIFAR-10 partitioned across five clients using Flower [6] for orchestration and Opacus [7] for local private training. We examine how non-IID data distributions, generated via symmetric Dirichlet partitioning, interact with DP-SGD mechanisms, and evaluate the reproducibility and numerical consistency of reported experimental artifacts.

Related Work. McMahan et al. [1] introduced federated averaging as a communication-efficient framework for decentralized deep learning. Abadi et al. [2] established DP-SGD, introducing the moments accountant for tight composition. Mironov [3] proposed Rényi Differential Privacy (RDP) as a relaxation that composes linearly under sequential training steps. Balle et al. [4] derived the optimal conversion theorem from RDP to (ε, δ)-DP implemented in Opacus.

Under static data partitions, parallel composition [5] bounds global privacy loss across disjoint client datasets. System orchestration is managed by Flower [6], while Opacus [7] handles per-sample clipping and noise injection.

2 Background

2.1 Federated Learning and FedAvg

FL coordinates model training over synchronized communication rounds $r \in \{1, \ldots, R\}$. At round r, a central server broadcasts the current global model parameters w^r to participating clients $k \in \{0, \ldots, K-1\}$. Each client trains locally on its local dataset $\mathcal{D}_k$ for E epochs using mini-batch stochastic gradient descent, yielding updated parameters w_k^{r+1}. The server aggregates the updates using weighted averaging:

$$w^{r+1} = \sum_{k=0}^{K-1} \frac{n_k}{N} w_k^{r+1}, \tag{1}$$

where $n_k = |\mathcal{D}_k|$ and $N = \sum_k n_k$.

2.2 Differential Privacy and DP-SGD

A randomized algorithm $\mathcal{M}$ satisfies (ε, δ)-Differential Privacy (DP) if, for all neighboring datasets $\mathcal{D}, \mathcal{D}'$ differing by the addition or removal of a single training example ($|\mathcal{D} \Delta \mathcal{D}'| = 1$, add/remove-one-sample adjacency), and all measurable output subsets $\mathcal{S}$:

$$\mathbb{P}[\mathcal{M}(\mathcal{D}) \in \mathcal{S}] \leq e^{\varepsilon} \mathbb{P}[\mathcal{M}(\mathcal{D}') \in \mathcal{S}] + \delta. \tag{2}$$

DP-SGD satisfies this guarantee by: (1) computing per-sample gradients $g_i(w) = \nabla_w \ell(w; x_i)$, (2) clipping their L_2 norms to a maximum threshold C, yielding $\bar{g}_i = g_i / \max(1, \|g_i\|_2 / C)$, and (3) adding calibrated Gaussian noise to the aggregated gradient. Because clipping bounds the contribution of any single example to $\|\bar{g}_i\|_2 \leq C$, adding or removing an example changes the summed batch gradient by at most C in L_2 norm, establishing global L_2 sensitivity $\Delta_2 = C$.

2.3 Rényi Differential Privacy (RDP)

Rényi Differential Privacy of order $\alpha_{\text{RDP}} > 1$ characterizes the Rényi divergence of order α_{RDP} between $\mathcal{M}(\mathcal{D})$ and $\mathcal{M}(\mathcal{D}')$. To prevent confusion throughout this report, we strictly denote the Rényi order as α_{RDP}, distinguishing it from the Dirichlet data heterogeneity parameter α. Under sequential composition, RDP values compose linearly: if client k performs T_k physical training steps, its cumulative RDP of order α_{RDP} is simply the sum of its step-level RDPs.

3 Threat Model and Privacy Scope

Sample-Level Privacy. The system targets **sample-level** differential privacy under add-or-remove-one-sample adjacency ($|\mathcal{D} \Delta \mathcal{D}'| = 1$, unbounded differential privacy). The reported privacy bounds quantify the statistical influence of adding or removing an individual CIFAR-10 training image within a client partition under the analytical Sampled Gaussian Mechanism in Opacus, where the per-step L_2 sensitivity of the summed clipped gradients is $\Delta_2 = C$. In contrast, substitute-one-sample adjacency (replacing an example, $|\mathcal{D} \Delta \mathcal{D}'| = 2$) would double the sensitivity to $2C$ or require group-privacy inflation, which is not supported by the accountant-reported ε values. All stated ε values represent accountant-reported estimates under the assumed Poisson-sampling mechanism rather than a proven end-to-end guarantee for the executed multi-round distributed system. The mechanism does **not** provide client-level differential privacy; it does not protect the collective dataset of a participating client.

Adversarial Capabilities. The threat model assumes an honest-but-curious central server and adversarial participants that observe shared model weights and round-level metrics. The server executes the aggregation protocol faithfully but may attempt to infer private training instances from received client parameters. Adversarial clients may observe global model checkpoints across rounds. Sample-level DP ensures that the presence or absence of any single training image has bounded statistical influence on the published models, mitigating membership inference and sample reconstruction attacks.

4 Methodology

4.1 DP-SGD Mechanism Implementation

Local training uses Opacus. For each mini-batch drawn with batch size B, the mechanism:

1. Computes per-sample gradients g_i for each example i in the batch.
2. Clips each gradient to maximum L_2 norm C: $\bar{g}_i = g_i / \max(1, \|g_i\|_2 / C)$.
3. Sums the clipped gradients: $\sum_i \bar{g}_i$.
4. Adds Gaussian noise with standard deviation σC to the summed clipped gradients:

$$\tilde{g} = \sum_{i=1}^{B} \bar{g}_i + \mathcal{N}(0, \sigma^2 C^2 I). \tag{3}$$

5. Normalizes the noisy sum by the expected batch size to obtain the descent step: $g_{\text{step}} = \tilde{g}/B$.

Because Gaussian noise standard deviation is scaled directly by C, the effective noise-to-sensitivity multiplier in the accountant is $\sigma = (\sigma C)/C$.

4.2 Privacy Accounting Chain

Privacy loss is tracked analytically via the subsampled Gaussian mechanism accountant in Opacus:

1. **Subsampling Rate**: At each local step, client k draws batches with empirical sampling rate $q_k = 1/\lceil n_k/B \rceil$, where $B = 64$ and n_k is the client's local sample count.
2. **Sequential RDP Composition**: Over T_k cumulative physical local steps across rounds, RDP accumulates linearly under adaptive composition:

$$\text{RDP}_{T_k}(\alpha_{\text{RDP}}) = T_k \cdot \text{RDP}_{\text{step}}(\alpha_{\text{RDP}}, q_k, \sigma). \tag{4}$$

3. **Conversion to (ε, δ)-DP**: Cumulative RDP is converted to (ε, δ)-DP at fixed $\delta = 10^{-5}$ via the optimal conversion theorem [4]:

$$\varepsilon_k(\delta) = \min_{\alpha_{\text{RDP}} > 1} \left[\text{RDP}_{T_k}(\alpha_{\text{RDP}}) - \frac{\log \delta + \log \alpha_{\text{RDP}}}{\alpha_{\text{RDP}} - 1} + \log \left(\frac{\alpha_{\text{RDP}} - 1}{\alpha_{\text{RDP}}} \right) \right]. \tag{5}$$

4. **Disjoint Partitions and Parallel Composition**: The full training dataset $\mathcal{D}$ is partitioned into pairwise disjoint client subsets $\mathcal{D}_0, \ldots, \mathcal{D}_{K-1}$ with $\mathcal{D}_j \cap \mathcal{D}_k = \emptyset$ for $j \neq k$. Under add/remove-one-sample adjacency, an added or removed training example affects the local dataset of exactly one client k^*, while all other client datasets $\mathcal{D}_j$ $(j \neq k^*)$ remain identical $(\mathcal{D}_j = \mathcal{D}'_j)$. At any communication round, only client k^* directly queries the partition

containing the sample; other clients $j \neq k^*$ access the sample solely through the broadcast global model parameters w^r, which acts as post-processing of earlier privatized updates. Client k^* adaptively composes its local DP-SGD queries over R rounds under RDP, and server-side FedAvg aggregation is a linear combination operating as post-processing. Consequently, under these modeling assumptions, the dataset-level privacy loss is bounded by the worst-case individual client:

$$\varepsilon_{\text{global}} = \max_{k \in \{0,\ldots,K-1\}} \varepsilon_k. \tag{6}$$

Under this formulation, the worst-case client is the client with the **smallest** local sample count n_k (Client 4 under $\alpha = 0.1$), because its higher sampling rate $q_k = 1/88$ drives faster privacy expenditure, yielding $\varepsilon_4 = 2.6974$. This global bound represents an accountant-reported analytical estimate under the assumed Poisson-sampling model; it relies on these composition assumptions and does not constitute a formal end-to-end mathematical proof of the full interactive system.

5 Experimental Setup

5.1 Hardware and Execution Environment

All reported experiments were executed on CPU using Ray single-machine virtual-client process simulation rather than a physical distributed deployment.

5.2 Model Architecture

We evaluate SimpleCNN with GroupNorm, designed to preserve sample independence (avoiding BatchNorm, which leaks cross-sample statistics):

- Three convolutional layers ($3 \to 16 \to 32 \to 64$ channels, 3×3 kernels, padding 1).
- GroupNorm with 4 groups following each convolutional layer.
- ReLU activations and 2×2 MaxPool layers.
- Fully connected layers ($64 \times 4 \times 4 \to 128 \to 10$) with Dropout ($p = 0.5$).

5.3 Data Partitioning

CIFAR-10 (50,000 training, 10,000 test images across 10 classes) is partitioned across 5 clients using a symmetric Dirichlet distribution parameterized by α:

- **Allocation vs. Class Support**: Individual images are assigned to disjoint client datasets; client datasets do not have disjoint classes, but rather class distributions overlap across clients while individual sample indices are strictly disjoint.
- $\alpha = 0.1$ **(High Heterogeneity)**: Client sample sizes are $[12944, 10514, 11934, 9029, 5579]$. The empirical heterogeneity metric (mean standard deviation of class counts across clients) is 1699.28.
- $\alpha = 10.0$ **(Near-IID)**: Client sample sizes are $[8856, 10009, 10553, 10399, 10183]$, with an empirical heterogeneity metric of 236.88.

Figure 1 displays the resulting class distributions under both partition settings.

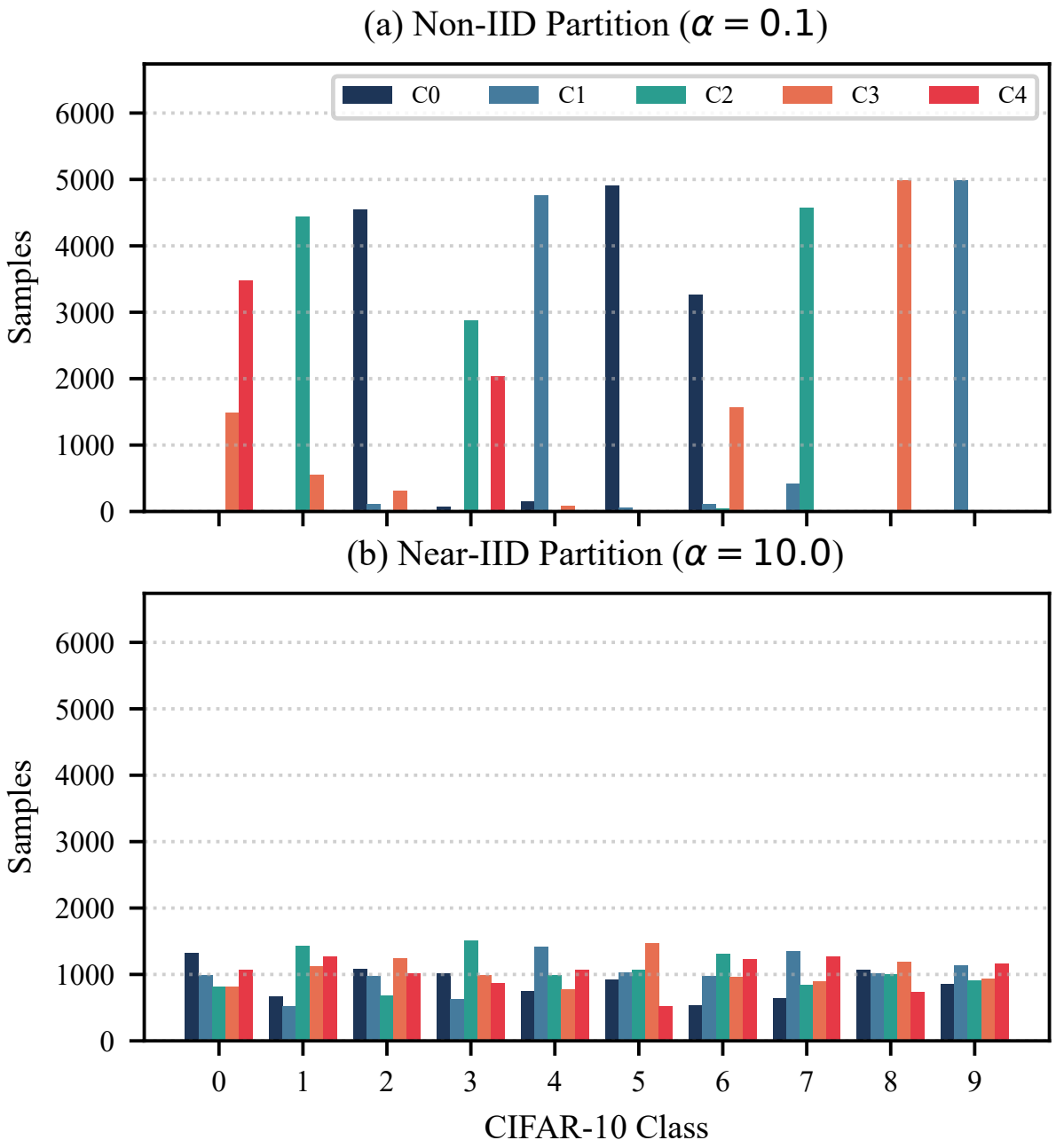


Figure 1: Dirichlet partition comparison across 5 clients: (a) high heterogeneity ($\alpha = 0.1$, metric 1699.28) and (b) near-IID ($\alpha = 10.0$, metric 236.88). Individual images are disjointly partitioned, while class distributions exhibit overlap.

5.4 Consolidated Hyperparameter Table

Table 1 consolidates all experimental hyperparameters across training stages. Where configurations differ, differences are explicitly recorded.

Hyperparameter	Centralized (S2)	FedAvg (S3)	DP-FL Grid (S4, S5)	Ablation (S6)
Dataset	CIFAR-10	CIFAR-10	CIFAR-10	CIFAR-10
Total Train Samples	50,000	50,000	50,000	50,000
Clients (K)	1 (Centralized)	5	5	5
Dirichlet Parameter (α)	N/A	0.1	0.1	0.1, 10.0
Communication Rounds (R)	N/A	15	15	15 (ref), 3 (multi-seed)
Local Epochs (E)	30 (epochs)	1	1	1
Batch Size (B)	64	64	64	64
Optimizer	Adam + CosineLR	SGD	SGD	SGD
Learning Rate	0.001	0.05	0.05	0.05
Momentum	$\beta_1 = 0.9, \beta_2 = 0.999$	0.0	0.9	0.9
Noise Multiplier (σ)	None	None	0.5, 1.0, 1.5, 2.0	1.0, 2.0
Clipping Bound (C)	None	None	0.1, 1.0	1.0
Target Delta (δ)	None	None	10^{-5}	10^{-5}
Execution Device	CPU	CPU	CPU	CPU
Random Seeds	42	42	42	42, 43, 44

Table 1: Consolidated hyperparameter configurations across experimental stages.

6 Results

6.1 Centralized and Non-Private FedAvg Baselines

The centralized baseline achieves **74.87%** test accuracy after 30 epochs of Adam optimization (learning rate 0.001). Non-private FedAvg achieves **47.38%** test accuracy after 15 communication rounds under $\alpha = 0.1$ data heterogeneity (SGD, learning rate 0.05, 1 local epoch per round).

Confounded Baseline Comparison. The 27.49 percentage-point difference between the centralized baseline (74.87%) and non-private FedAvg (47.38%) does not represent the cost of differential privacy or data heterogeneity alone. The comparison is confounded by optimizer selection (Adam with cosine annealing vs. fixed SGD), learning rate (0.001 vs. 0.05), training budget (30 full epochs vs. 15 federated rounds with 1 local epoch), and federated aggregation dynamics. No near-IID non-private FedAvg baseline was established in the experimental records, precluding causal isolation of these individual factors.

6.2 Main DP-FL Result (Stage 4)

The primary Stage 4 DP-FL configuration ($\alpha = 0.1, \sigma = 1.0, C = 1.0$, seed 42) attains a final test accuracy of **26.11%** (with a best round accuracy of 27.29% at round 14) and an accountant-reported cumulative privacy bound of $\varepsilon = 2.6974$ ($\delta = 10^{-5}$, minimizing Rényi order $\alpha_{\text{RDP}} = 6.9$).

Figure 2 shows test accuracy convergence over 15 communication rounds alongside the centralized baseline reference line (74.87%). Figure 3 presents the corresponding training loss trajectory. Figure 4 depicts cumulative privacy expenditure (ε).

6.3 Noise Multiplier and Clipping Threshold Grid (Stage 5)

Table 2 reports the authoritative 15-round Stage 5 grid search across noise multipliers $\sigma \in \{0.5, 1.0, 1.5, 2.0\}$ and clipping bounds $C \in \{0.1, 1.0\}$ under $\alpha = 0.1$ and seed 42.

Non-Monotonicity and Clipping Dynamics. The grid results refute the blanket assumption that increasing σ uniformly degrades accuracy across all configurations:

- For $C = 0.1$, final accuracy remains remarkably stable across noise scales: 30.61% ($\sigma = 0.5$), 32.01% ($\sigma = 1.0$), 32.06% ($\sigma = 1.5$), and 30.57% ($\sigma = 2.0$). In this grid, configuring

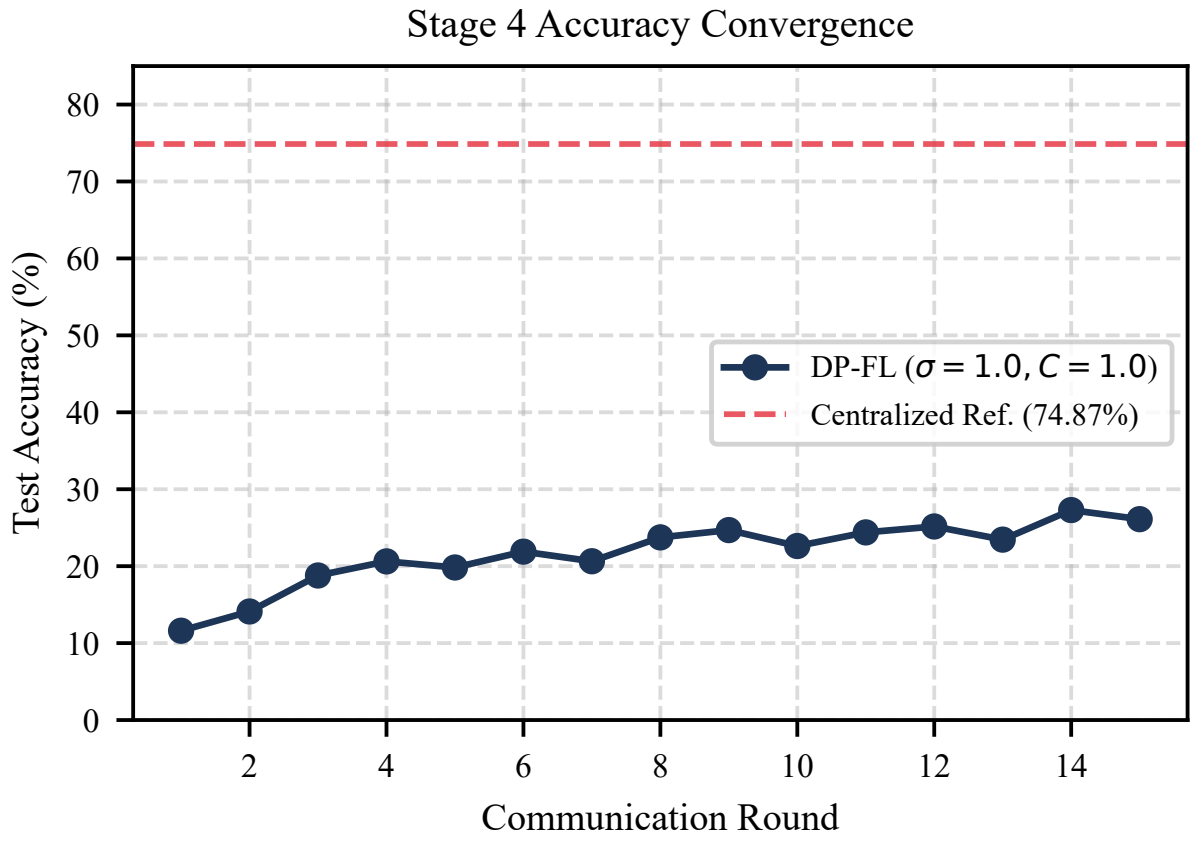


Figure 2: Stage 4 test accuracy convergence over 15 communication rounds ($\alpha = 0.1, \sigma = 1.0, C = 1.0$, seed 42) with the centralized baseline (74.87%) shown as a horizontal reference line.

Noise Multiplier (σ)	**Accountant Epsilon** (ε **at** $\delta = 10^{-5}$)	**Accuracy at** $C = 0.1$	**Accuracy at** $C = 1.0$
0.5	19.0778	30.61%	36.49%
1.0	2.6974	32.01%	28.74%
1.5	1.3436	32.06%	17.27%
2.0	0.9117	30.57%	18.37%

Table 2: Stage 5 privacy–utility grid across 15 communication rounds ($\alpha = 0.1$, seed 42). Accountant-reported epsilon is identical across clipping thresholds C for a given σ because C scales both clipping sensitivity and Gaussian noise standard deviation (σC), leaving the noise multiplier relative to sensitivity invariant in the accountant.

$\sigma = 2.0$ with $C = 0.1$ attains **30.57%** test accuracy at a formal bound of $\varepsilon = 0.9117$. However, this is an empirical observation from a single training run (seed 42).

- Conversely, for $C = 1.0$, accuracy drops steeply as noise increases: from 36.49% ($\sigma = 0.5$) down to 28.74% ($\sigma = 1.0$), 17.27% ($\sigma = 1.5$), and 18.37% ($\sigma = 2.0$).

Figure 5 illustrates these distinct trajectories. The observed results demonstrate that formal privacy bounds below $\varepsilon < 2.0$ do not mathematically prevent accuracy from reaching 30%, but they do not establish a causal effect of clipping threshold C independently of effective step size and gradient signal-to-noise ratio.

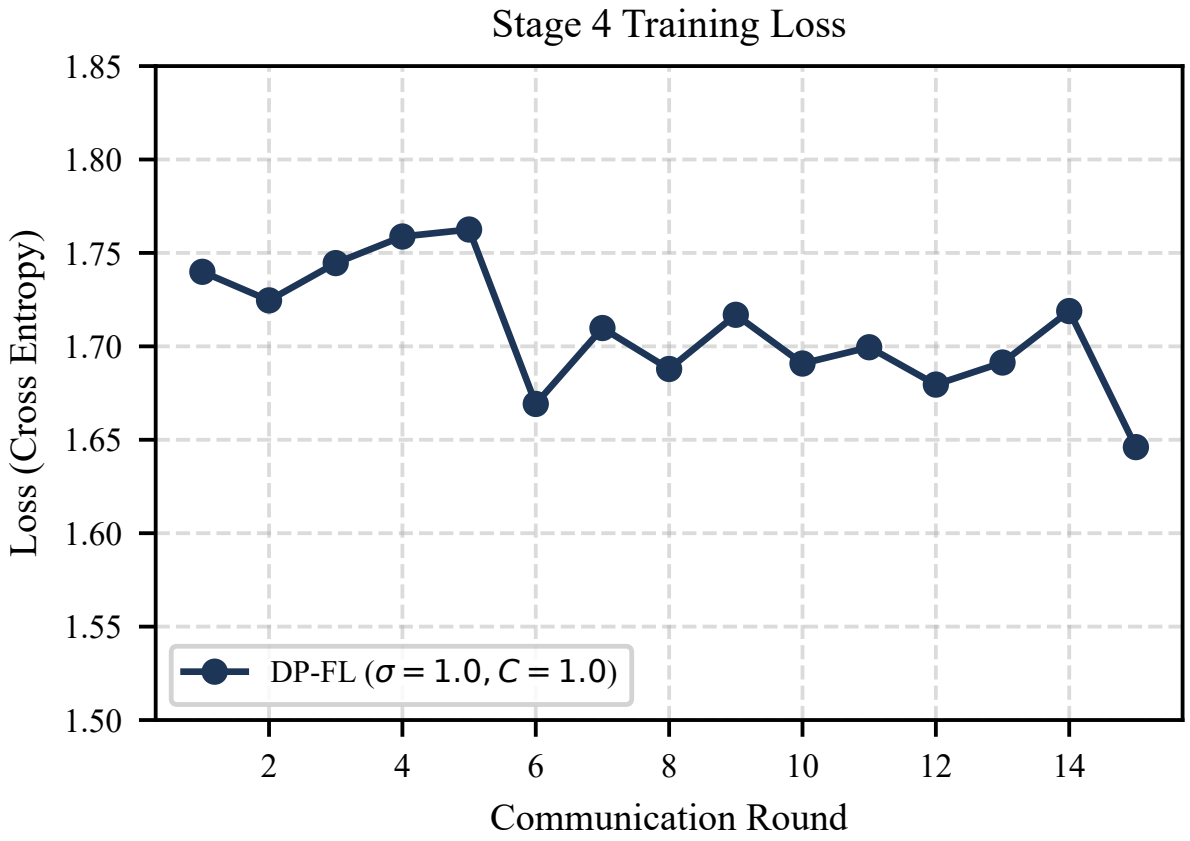


Figure 3: Stage 4 training loss trajectory across 15 communication rounds.

6.4 Data Heterogeneity Ablation (Stage 6)

Table 3 summarizes the authoritative 15-round Stage 6 reference runs comparing high skew ($\alpha = 0.1$) against near-IID data ($\alpha = 10.0$).

α	σ	C	Seed	Best Order α_{RDP}	ε ($\delta = 10^{-5}$)	Final Accuracy
0.1	1.0	1.0	42	6.9	2.6974	24.41%
10.0	1.0	1.0	42	8.3	2.0691	31.51%
0.1	2.0	1.0	42	19.0	0.9117	13.79%
10.0	2.0	1.0	44	23.0	0.7271	15.26%

Table 3: Stage 6 15-round data heterogeneity ablation runs on full CIFAR-10 across 5 clients.

Experimental Observations.

- At $\sigma = 1.0$ (seed 42), the near-IID run achieves 31.51% final accuracy compared to 24.41% under high heterogeneity ($\alpha = 0.1$). Both values represent single-run outcomes from seed 42. In light of the observed 4.33 percentage-point spread across nominally identical runs, this single-seed comparison cannot be causally attributed to data heterogeneity alone.
- At $\sigma = 2.0$, the near-IID run used seed 44 (15.26%) while the $\alpha = 0.1$ run used seed 42 (13.79%), due to an IPC worker restart on seed 42. Because random seeds differ, this is a non-like-for-like comparison and cannot be attributed to data heterogeneity alone.

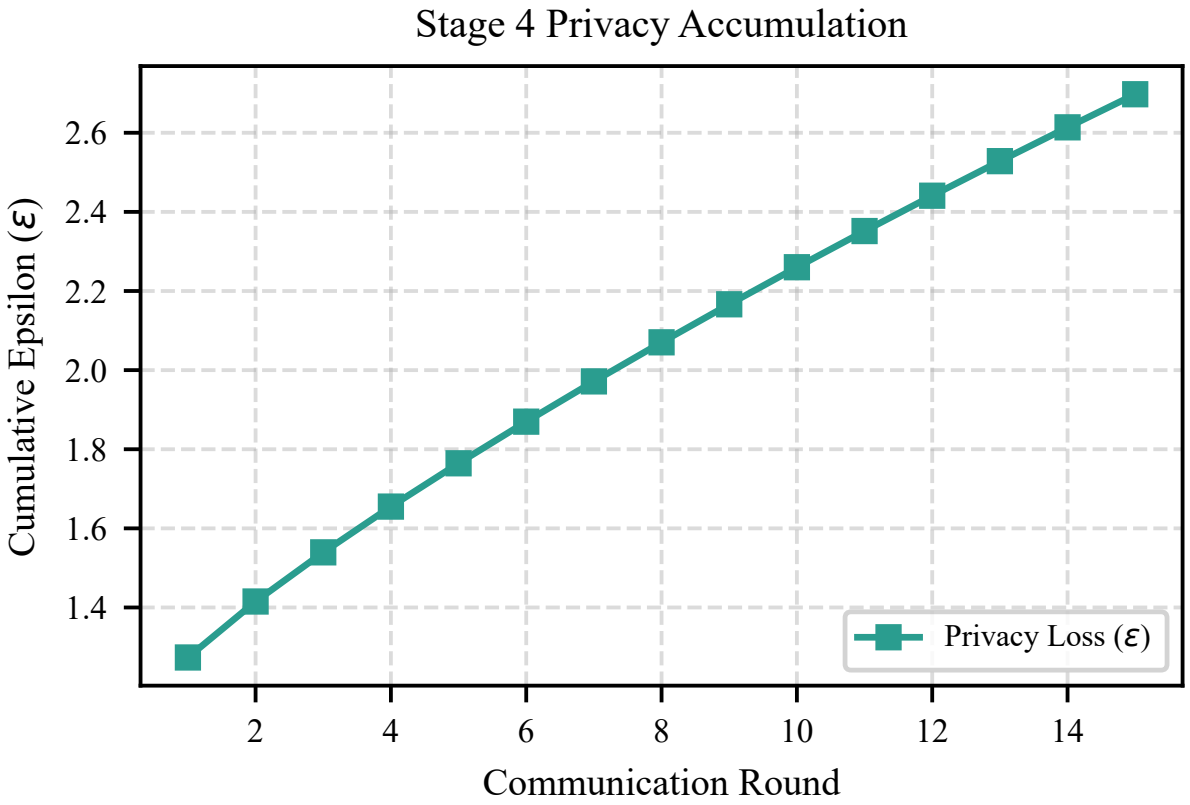


Figure 4: Stage 4 cumulative epsilon (ε vs. communication round) under RDP accounting at $\delta = 10^{-5}$.

- Near-IID partitioning yields lower cumulative epsilon ($\varepsilon = 2.0691$ vs. 2.6974 at $\sigma = 1.0$) because client sample sizes are more evenly balanced ($n_k \approx 10,000$), reducing the worst-case sampling rate q_k.

Figures 6 and 7 display the Stage 6 ablation trajectories and paired comparisons.

6.5 Multi-Seed Robustness Evaluation

To evaluate sensitivity to random seed initialization, multi-seed experiments were performed across three seeds ($\{42, 43, 44\}$, $N = 3$) over 3 communication rounds on full CIFAR-10. Table 4 presents empirical means and sample standard deviations (s, computed with Bessel's correction ddof $= 1$).

α	σ	Seeds	Final Acc. Mean	Final Acc. Std (s)	Epsilon Mean	Epsilon Std (s)
0.1	1.0	42, 43, 44	18.66%	±0.93%	1.5171	±0.0215
0.1	2.0	42, 43, 44	10.96%	±1.07%	0.3712	±0.0301
10.0	1.0	42, 43, 44	17.88%	±2.36%	1.2354	±0.0214
10.0	2.0	42, 43, 44	9.77%	±0.69%	0.3073	±0.0133

Table 4: Multi-seed empirical robustness summary ($N = 3$ seeds, 3 communication rounds). Standard deviations denote sample standard deviations (s with ddof $= 1$). Underlying pipeline files record population standard deviations (ddof $= 0$).

Interpretation of Multi-Seed Results. These experiments cover only three communication rounds and produce low early accuracies (9.77%–18.66%). The observed accuracy differences

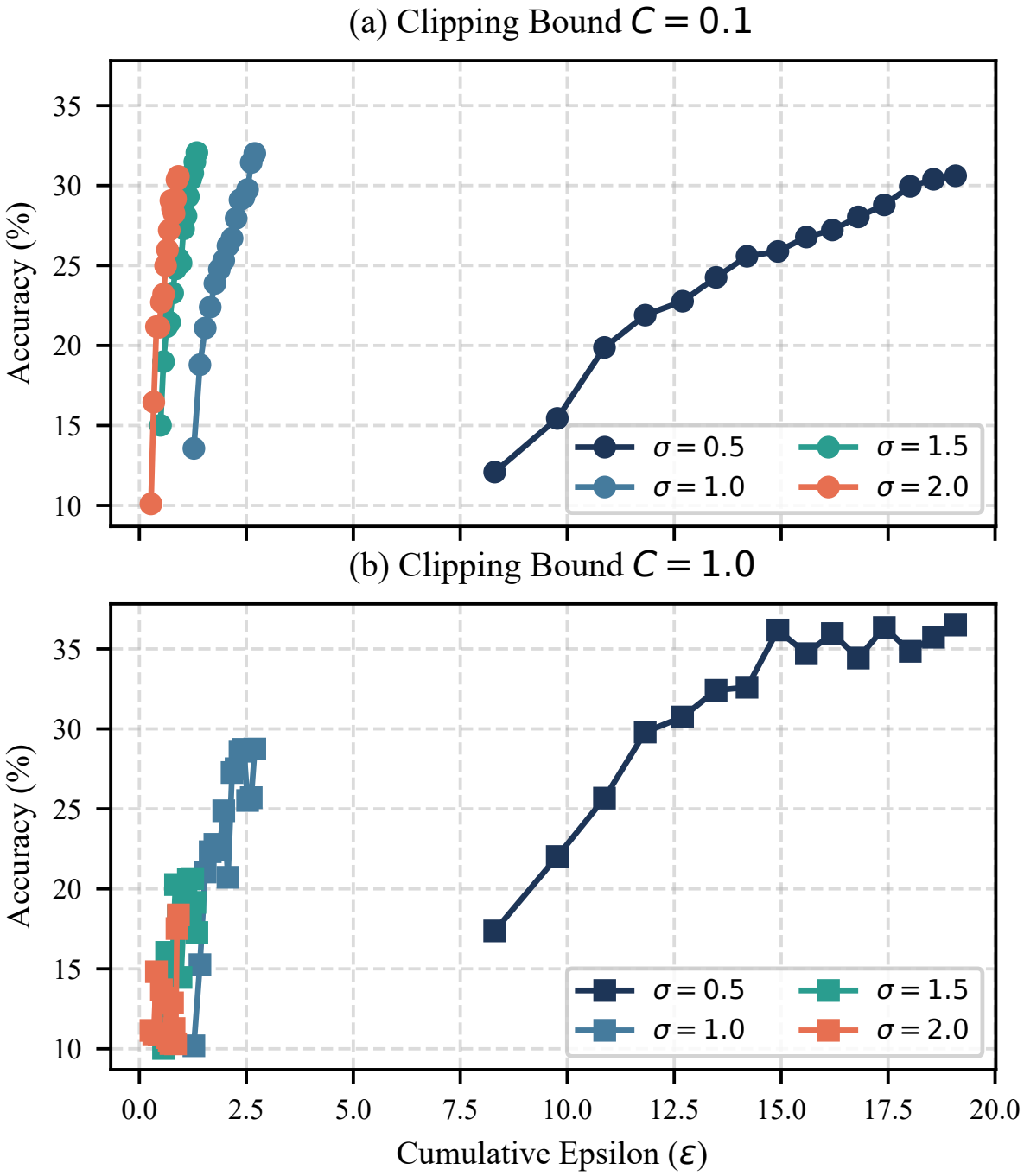


Figure 5: Stage 5 privacy–utility tradeoff across 15 rounds for (a) $C = 0.1$ and (b) $C = 1.0$.

between $\alpha = 0.1$ and $\alpha = 10.0$ (+0.78% at $\sigma = 1.0$ and +1.19% at $\sigma = 2.0$ in favor of $\alpha = 0.1$) fall well within sample standard deviations (±2.36% and ±1.07%). They provide preliminary evidence of seed variability, but do not provide evidence for or against a general data heterogeneity effect over full training horizons. We make no claim of statistical significance or equivalence.

Figure 8 plots the 3-round multi-seed summary against the 15-round reference runs, and Figure 9 illustrates the combined tradeoff across Stages 5 and 6.

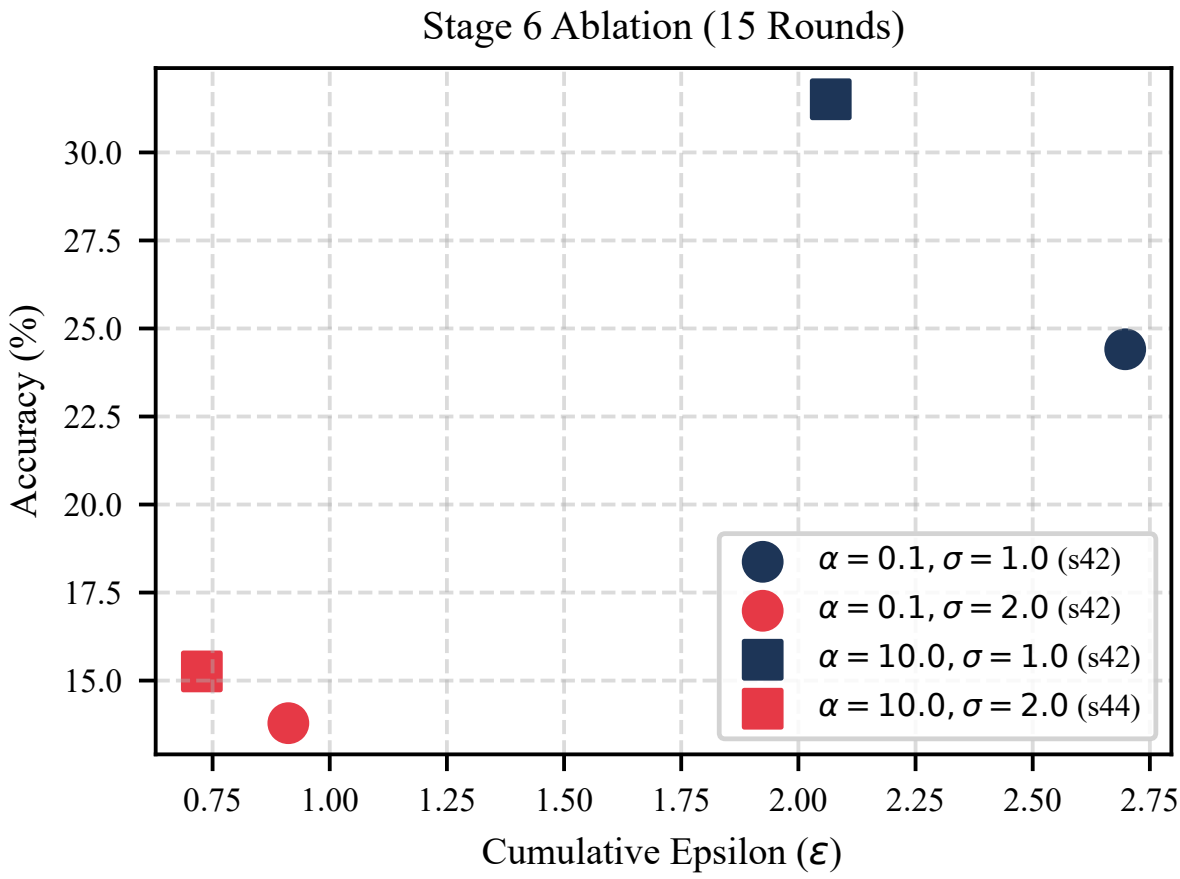


Figure 6: Stage 6 15-round ablation results comparing $\alpha = 0.1$ and $\alpha = 10.0$ under $\sigma = 1.0$ and $\sigma = 2.0$.

6.6 Evaluation of Nominally Identical Historical Runs

During artifact inspection, three distinct final test accuracies were identified for nominally identical configurations ($\alpha = 0.1, \sigma = 1.0, C = 1.0$, seed 42, 15 rounds, 5 clients, SGD learning rate 0.05, batch size 64). Table 5 compares these runs against source artifacts.

Experimental Stage	Final Acc.	Best Acc.	Accountant ε ($\delta = 10^{-5}$)	Execution Date	Source Artifact
Stage 4 DP-FL	26.11%	27.29%	2.6974	October 7, 2026	`stage4/summary.json`
Stage 5 Grid	28.74%	28.74%	2.6974	October 9, 2026	`stage5/.../summary.json`
Stage 6 Ablation	24.41%	27.14%	2.6974	October 9, 2026	`stage6/.../summary.json`

Table 5: Comparison of historical 15-round runs sharing identical nominal hyperparameters ($\alpha = 0.1, \sigma = 1.0, C = 1.0$, seed 42). All runs predate the determinism patch; reported ε values reflect accountant estimates under assumed Poisson subsampling.

Evaluation Findings.

1. **Accounting Agreement**: All three runs exhibit identical privacy accounting metrics: $\varepsilon = 2.6974$ at $\delta = 10^{-5}$, minimizing Rényi order $\alpha_{\text{RDP}} = 6.9$, and client 4 governing the global bound ($T_4 = 1320$ steps, $q_4 = 1/88$).

2. **Observed Accuracy Spread**: Across the three executions, final test accuracy ranges from 24.41% to 28.74%, representing an observed spread of approximately 4.33 percentage points. This spread reflects empirical variability across historical runs conducted before

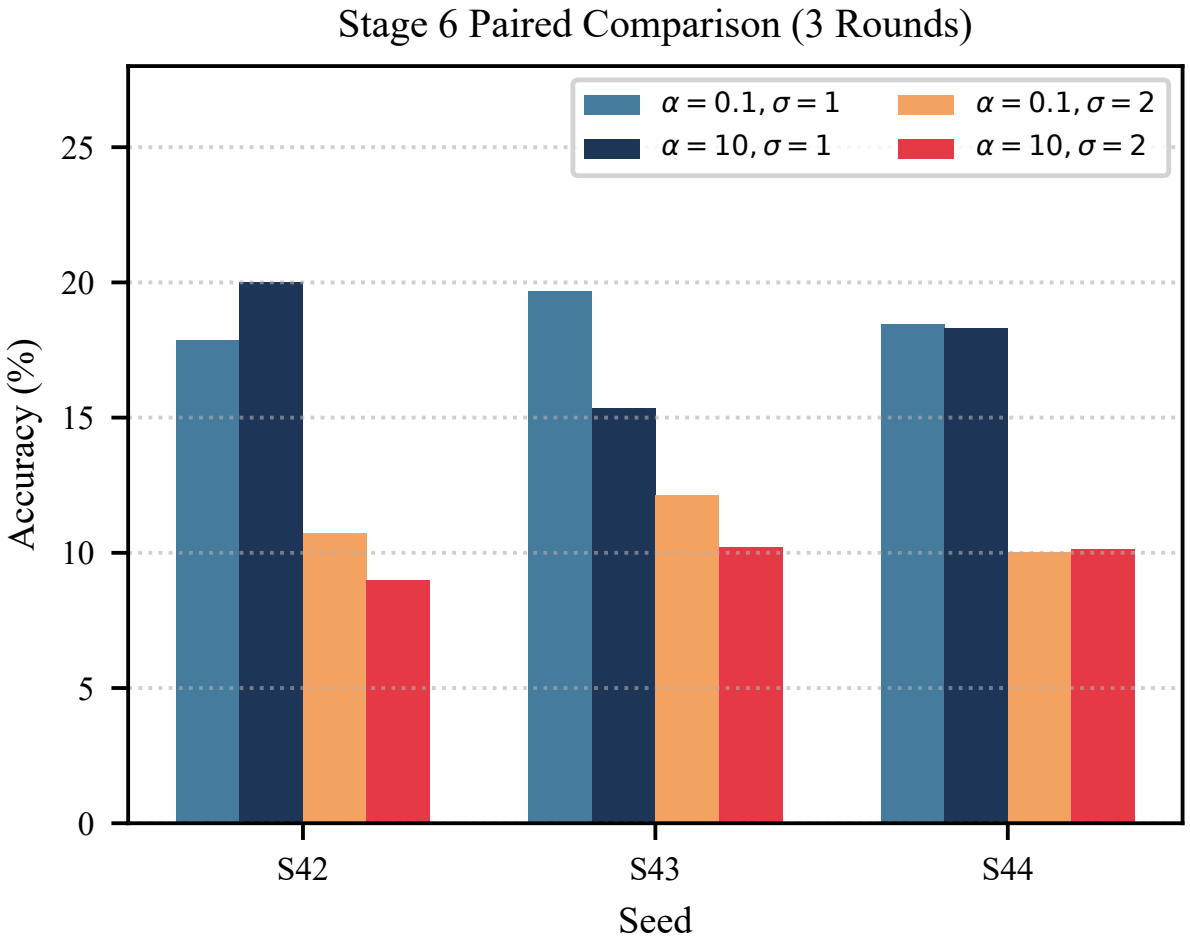


Figure 7: Stage 6 paired comparison across random seeds 42, 43, and 44 over 3 communication rounds.

deterministic seeding was formalized across client worker processes; it is an observed sample range, not a formal noise floor or statistical confidence interval.

3. **Historical Non-Determinism**: All three executions predated the repository's deterministic seeding architecture (detailed in Appendix B). Prior to that fix, the federated pipeline lacked per-round client seed derivation, isolated DataLoader RNG generators, and deterministic server aggregation sorting across Ray actor processes. In addition, minor runner script refactorings occurred across stages. Consequently, identical nominal seeds did not yield identical historical trajectories.

4. **Attribution and Reproducibility**: Because multiple software factors evolved across early stages, the exact causal attribution of accuracy differences cannot be conclusively isolated from historical artifacts alone. Historical runs cannot be reproduced bitwise; determinism has been demonstrated solely on post-patch integration tests (Appendix B).

7 Discussion

Privacy Budget vs. Model Utility. Our empirical findings demonstrate that formal privacy bounds do not map monotonically to classification accuracy. Prior assumptions in DP-FL often suggest that enforcing $\varepsilon < 2.0$ necessarily constrains accuracy below 30%. However:

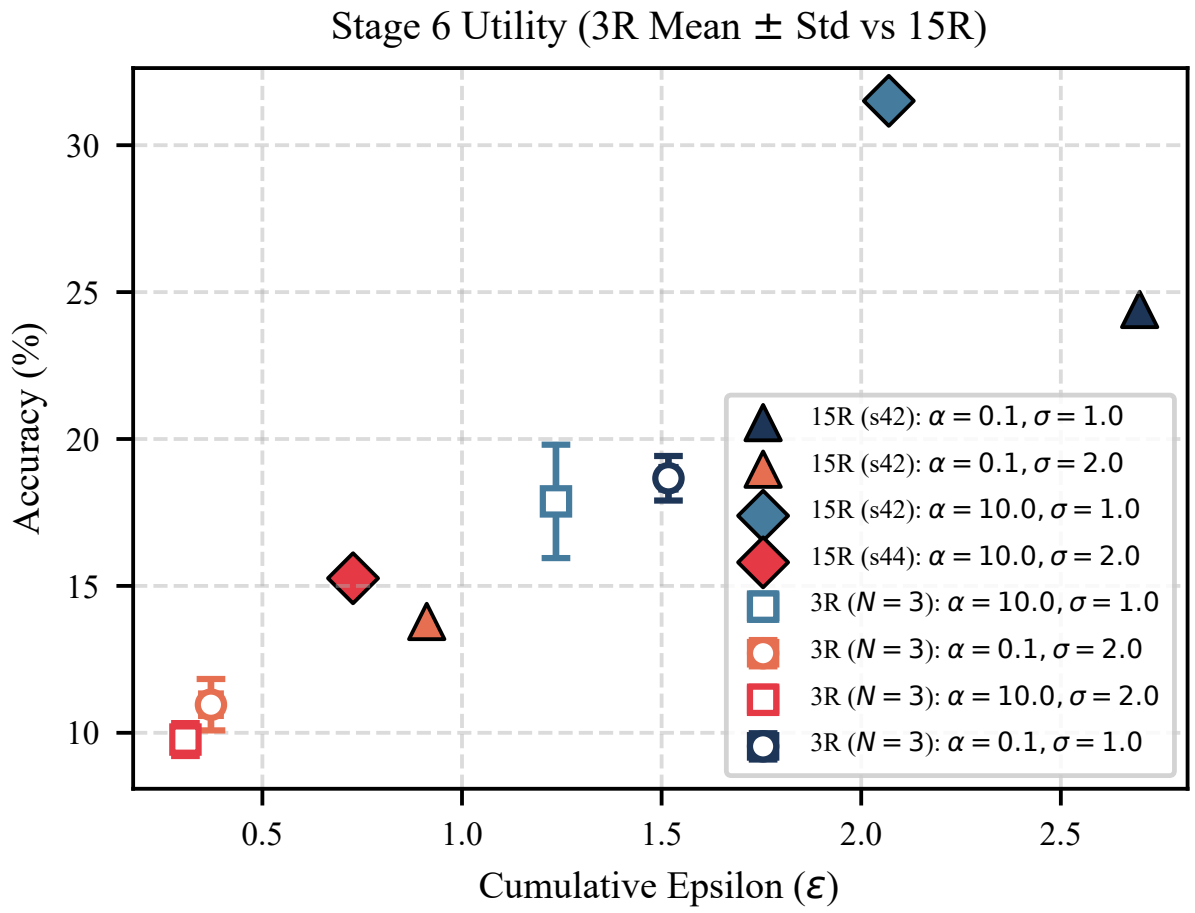


Figure 8: Stage 6 multi-seed summary (3-round mean ± sample standard deviation) alongside 15-round reference runs. Distinct markers and shared colors differentiate 3-round from 15-round series.

- In Stage 5, configuring $\sigma = 2.0$ with $C = 0.1$ achieved **30.57%** test accuracy at $\varepsilon = 0.9117$ in a single run.
- In Stage 6, the near-IID configuration ($\alpha = 10.0, \sigma = 1.0, C = 1.0$) achieved **31.51%** accuracy at $\varepsilon = 2.0691$ in a single run.

Model utility depends intricately on gradient clipping norm C, local gradient variance, and optimization dynamics, rather than differential privacy budget ε in isolation. Moreover, the observed results do not establish a causal effect of clipping threshold C independently of other training factors. Clipping interacts directly with effective learning rate, gradient signal-to-noise ratio, and batch variance; varying C without retuning the learning rate simultaneously shifts both the optimization step size and the injected noise scale.

Finite-Horizon Evaluation. The 15-round runs provide empirical measurements over a 15-round horizon on CPU. They do not demonstrate asymptotic convergence. Extended training horizons and systematic learning rate scheduling would be required to evaluate long-term convergence behavior.

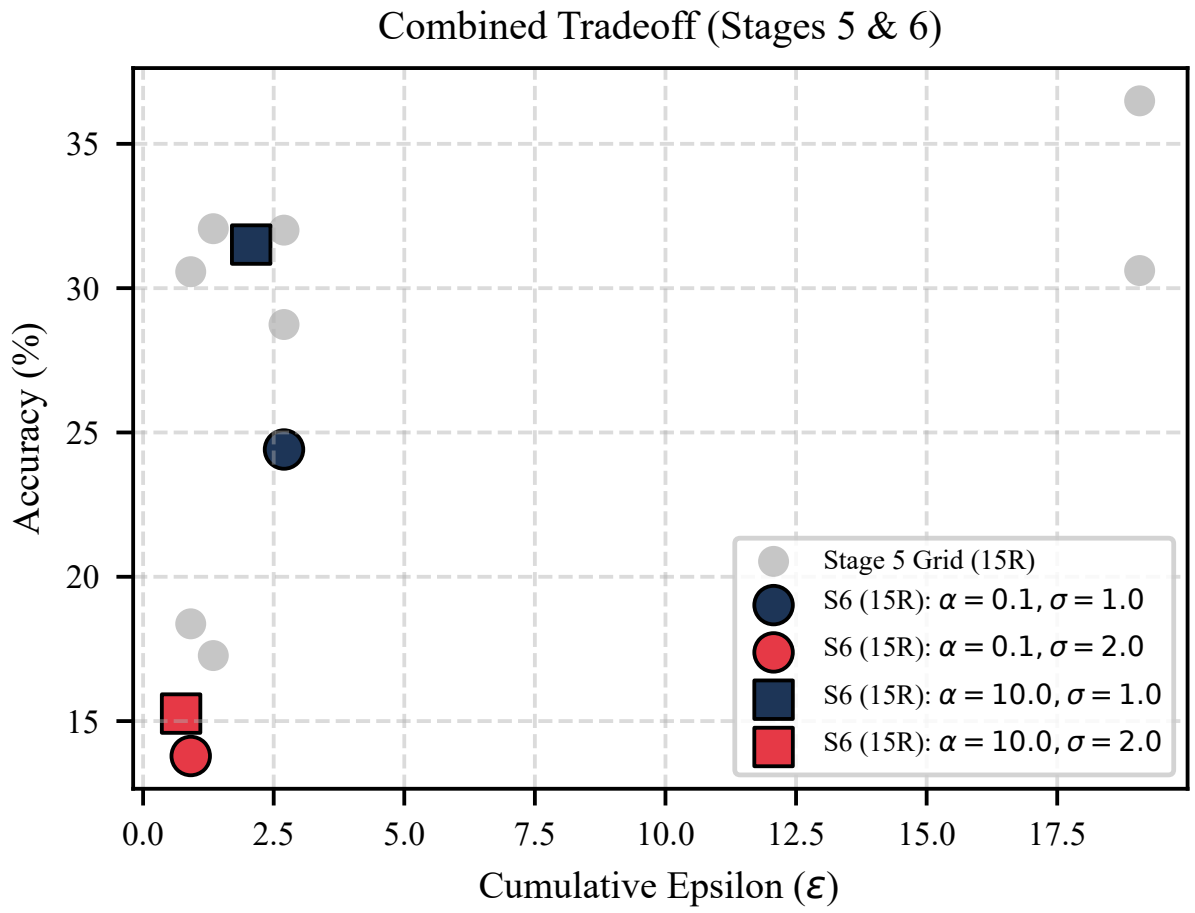


Figure 9: Combined privacy–utility tradeoff across Stage 5 grid search and Stage 6 ablation runs.

8 Limitations

1. **Privacy Scope and Adjacency**: The privacy guarantee strictly targets individual CIFAR-10 training images under add-or-remove-one-sample adjacency (sample-level DP), not entire participating clients or institutions.

2. **Sampling Mechanism and Accountant Assumptions**: Poisson subsampling denotes independent per-example inclusion (Bernoulli trials with probability q), not sampling with replacement. Although Opacus wraps the loader in its Poisson sampler (`UniformWithReplacementSampler`, executing independent Bernoulli inclusion per step), the stated ε values represent accountant-reported estimates under the assumed Poisson-sampling mechanism. The end-to-end formal guarantee for the executed multi-round sampling procedure and model-sharing protocol has not been established.

3. **Multi-Round Composition and Verification**: Global privacy bounds rely on disjoint client data partitions and post-processing across communication rounds. This reasoning assumes the accountant's analytical model applies without side channels; it has not been verified by an empirical privacy audit (such as membership inference attacks or canary gradient reconstruction) or a formal external mathematical proof.

4. **Secure Random Number Generation**: For simulation efficiency, Opacus cryptographically secure RNG was disabled (`secure_rng=False`). Production deployments requiring formal differential privacy must enable cryptographically secure RNG.

5. **Hyperparameter Tuning and Causality**: Federated learning rates and momentum were kept fixed across DP runs without DP-specific tuning due to CPU computational constraints. Observed performance differences between clipping thresholds cannot be interpreted as isolated causal effects.

6. **Multi-Seed Horizon**: Multi-seed runs were limited to 3 communication rounds and 3 seeds, serving as preliminary variance checks rather than asymptotic proofs.

9 Conclusion

We conducted an empirical evaluation and reproducibility analysis of Differentially Private Federated Learning on CIFAR-10 across 5 clients. On full CIFAR-10, centralized non-private training reaches 74.87% accuracy (Adam, 30 epochs), non-private FedAvg achieves 47.38% accuracy (SGD, 15 rounds), and DP-FL attains 26.11% accuracy at an accountant-reported estimate of $\varepsilon = 2.6974$ under assumed Poisson subsampling ($\sigma = 1.0, C = 1.0$, 15 rounds). Across three historical runs with nominally identical configurations, final accuracy ranged from 24.41% to 28.74%, an observed spread of approximately 4.33 percentage points. Grid evaluations demonstrated that $\sigma = 2.0$ with $C = 0.1$ attained 30.57% accuracy at an accountant-reported $\varepsilon = 0.9117$ in a single run, though this does not isolate a causal effect of clipping threshold C. Direct baseline comparisons remain confounded by optimizer choice and training budgets. These findings emphasize the necessity of rigorous empirical accounting, transparent provenance, and disciplined qualification of variability in private federated learning systems.

Appendix

A Detailed Privacy Accounting Breakdown

Tables 6 and 7 record the verified per-client accounting breakdown for $\sigma = 1.0, C = 1.0$ under $\alpha = 0.1$ and $\alpha = 10.0$ across 15 communication rounds.

Client k	**Samples** n_k	**Sample Rate** q_k	**Steps** T_k	**Epsilon** ε_k	**Minimizing Rényi Order**
0	12,944	0.00493	3,045	1.6808	9.4
1	10,514	0.00606	2,475	1.8797	8.8
2	11,934	0.00535	2,805	1.7560	9.1
3	9,029	0.00704	2,130	2.0442	8.4
4	5,579	0.01136	1,320	2.6974	6.9

Table 6: Per-client accounting breakdown for $\alpha = 0.1, \sigma = 1.0, C = 1.0$, seed 42 (15 rounds). Client 4 determines global privacy ($\varepsilon_{\text{global}} = 2.6974$) due to its highest sampling rate, while Client 0 executed the most physical steps ($T_0 = 3,045$).

Client k	**Samples** n_k	**Sample Rate** q_k	**Steps** T_k	**Epsilon** ε_k	**Minimizing Rényi Order**
0	8,856	0.00719	2,085	2.0691	8.3
1	10,009	0.00637	2,355	1.9323	8.7
2	10,553	0.00606	2,475	1.8797	8.8
3	10,399	0.00613	2,445	1.8925	8.8
4	10,183	0.00625	2,400	1.9120	8.7

Table 7: Per-client accounting breakdown for $\alpha = 10.0, \sigma = 1.0, C = 1.0$, seed 42 (15 rounds). Client 0 determines global privacy ($\varepsilon_{\text{global}} = 2.0691$), while Client 2 executed the most physical steps ($T_2 = 2,475$).

B Reproducibility Architecture and Verification

B.1 Deterministic Seeding Architecture

Prior to commit `08ad3f8` on October 11, 2026, experimental executions across Stages 4, 5, and 6 (conducted between October 7 and October 9, 2026) lacked deterministic worker seeding, per-round client seed derivation, and isolated DataLoader RNG generators. This historical non-determinism accounts for the observed 4.33 percentage-point spread across nominally identical runs. In commit `08ad3f8`, determinism was formalized in `run_federated.py`:

- **Root Initialization**: Seeds Python `random`, NumPy, and PyTorch RNGs using base seed.
- **Per-Client Seeding**: Deterministically derives an integer seed via SHA-256:

$$\text{seed}_{k,r} = \text{int}\Big(\text{SHA256}(s_{\text{base}} : k : r)[:4]\Big) \;\&\; \text{0x7FFFFFFF}. \tag{7}$$

- **Isolated Generator**: Passes an isolated `torch.Generator` to DataLoader shuffling and Opacus `make_private`.
- **Deterministic Initial Model**: Initializes the initial global model from a seeded instance and passes it to the server strategy as `initial_parameters`.
- **Deterministic Aggregation**: In `CheckpointingFedAvg.aggregate_fit()`, incoming client results are sorted deterministically by numeric client ID before averaging.

B.2 Integration Verification Artifacts

Reproducibility was verified across two-round integration runs in `results/reproducibility/`:

- **Run A vs. Run B (Seed 42)**: Both runs produced identical initial model hash (`00c43041...`), identical final model hash (`e23b740f...`), identical cumulative epsilon ($\varepsilon = 4.442963$), identical best Rényi order (4.1), identical total DP steps (30), and identical per-round accuracy (13.5% round 1, 12.5% round 2).
- **Divergence Check (Seed 43)**: Seed 43 produced distinct initial and final model hashes, different accuracy (7.5%), and different epsilon ($\varepsilon = 4.696514$), while maintaining structural step count agreement (30 total DP steps).

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